Delegating Investment Strategy to AI Agents: Performance Measurement and Risk in a Human-Benchmarked Tournament*

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Abstract

We propose a simulation methodology for building and evaluating investment strategies and apply it to judging whether to delegate to machines. Autonomous AI agents build strategies across many paths from a market simulator calibrated to real-world merger arbitrage. Each strategy is then locked and evaluated on test paths also traded by human competitors. We avoid look-ahead bias, overfitting from data snooping, and lack of robustness to the *distribution* of potential outcomes. Practice rank predicts out-of-sample rank, yet the *average* agent substantially underperforms the human median. The gap measures what human guidance, supervision, and governance still add to autonomous agents.

Keywords: autonomous AI systems as delegated traders and strategy developers; a simulation methodology for building and evaluating investment strategies; AI and machine learning benchmarked to human performance; tournament incentives; merger-arbitrage strategies; risk management

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1 Introduction

An important frontier for agentic artificial intelligence in finance is moving from prediction to delegation. A language model can be asked whether a headline is positive or negative, but an autonomous financial agent must do considerably more. It must translate a mandate into a research process, decide what evidence to collect, write an executable policy, validate that policy against noisy outcomes, allocate scarce development time, and determine how much capital to allocate and how much tail risk to accept. Once the autonomous system is allowed to revise its own procedure and submit orders without contemporaneous human approval, the central economic problem is no longer text classification. It is the familiar problem of selecting and governing a delegated investment manager, now applied to a machine delegate.

How should a principal decide whether such a delegate is ready? The usual evidence, a backtest on historical market data or a live track record, is poorly suited to the question. History supplies a single realized path, and that path is only one draw from the distribution of things that could have happened. A strategy developed and evaluated on the same path will be fitted to its particular news, deal outcomes, and price moves, so the reported performance overstates what the strategy would earn on the next path (Lo and MacKinlay, 1990; Sullivan, Timmermann, and White, 1999; Bailey, Borwein, López de Prado, and Zhu, 2014). Statistical corrections for data snooping (White, 2000; Hansen, 2005; Romano and Wolf, 2005; Harvey, Liu, and Zhu, 2016) adjust for the number of strategies tried, but take the single path as given: they require a count of trials that is rarely recorded, and they cannot supply adverse events that the path did not contain. For language-model agents there is an additional problem: the historical evaluation period may overlap the model's training data, so apparent foresight can be memory (Glasserman and Lin, 2023). Comparisons of live returns add their own confounding variables. Different systems and different people act on different information, under different market conditions, mandates, capital, account limits, and timing, and with implementation skills that are rarely observed. Realized returns combine strategy quality with the luck of the path.

Our methodology is motivated by simulation-based learning, in which participants build and test decision strategies in a calibrated market simulator before they are evaluated (Mak

and McCurdy, 2019). Aviation faced an analogous problem and solved it with simulators. A trainee pilot is not certified on the strength of the one flight that happened to go smoothly, and is not trained by flying a plane full of people, potentially into a storm. Pilots practise in flight simulators that present the range of routine and unusual situations they are likely to encounter, including rare emergencies and cascading failures in the tail of the distribution. They are debriefed and practise again, and eventually tested on scenarios they have not rehearsed before they are trusted with passengers. The simulator makes it possible both to *build* skills that are robust to what could happen and to *evaluate* those skills cleanly before deployment.

This paper applies the same two-stage logic to a delegated investment strategy and uses it to compare autonomous AI agents with human decision makers from a highly competitive international competition. In the first stage, participants derive their best strategy. AI agents and human competitors receive the same mandate, information, and account settings, and prepare over the same sequence of build intervals followed by scored practice sessions, each of which evaluates the current strategy and feeds into the next round of revision. Practice paths are generated by a market simulator whose stochastic environment (deal outcomes, news arrivals, and price dynamics) is calibrated to real-world merger arbitrage. The build phase therefore offers a *Monte Carlo view of history*: rather than tuning a strategy to the one historical path that did occur, participants confront many paths that could have occurred, including adverse and unusual ones, and adapt their models to be robust to the typical range of outcomes.

In the second stage, each agent's best strategy is locked and applied with no further revision to ten test (out-of-sample) paths of the market containing new, previously unseen information. These paths are generated by projecting the calibrated distribution forward, so they represent the range of conditions the locked strategy could face after deployment rather than a reserved segment of the one path that occurred. Because the strategy is fixed before any of these paths are revealed, its measured performance does not include look-ahead bias from model development or training. Because the evaluation spans ten paths rather than one, it reflects a range of potential out-of-sample realizations rather than a single draw. And because those paths are saved and replayed, the AI agents face exactly the news sequences,

price paths, and account limits that the human participants faced in the competition, so every performance ranking compares the two groups on identical realizations. The control the simulator provides (a common mandate and feedback, the same news sequences and price paths, replayable test paths with the same stochastics for agents and humans, identical account limits and preparation time, and a complete archive of every decision) removes the confounding variables that make performance comparisons between AI and human investors less informative when they rely on market data.

Two further features make the comparison economically meaningful. First, the human participants act as principals in the sense that they may use any tools available to them, including AI, but they prepare on the same schedule of build intervals and scored practice sessions as the autonomous agents. The comparison therefore asks whether AI agents can operate autonomously or whether they still perform better with human guidance, supervision, and governance. Second, incentives replicate the profit motive that is often missing from simulation studies. The human participants compete for prizes and standing at a global competition, a motivation that is as strong as managing a portfolio for profit, while the AI agents are directed to perform as well as possible and, at each practice session, observe their own results alongside the human leaderboard on the same realizations and rebuild their strategies to become more competitive.

Our application is merger arbitrage, where competence at one stage of the decision need not survive the stages that follow. Financial text contains economically meaningful information: Loughran and McDonald (2011) show why domain-specific language matters, and Jeon, McCurdy, and Zhao (2022) connect identifiable news arrivals to jumps in individual stock returns. Large language models (LLMs) can extract return-relevant content from news (Lopez-Lira and Tang, 2026), and asset managers increasingly employ generative AI in practice (Sheng, Sun, Yang, and Zhang, 2026). Yet a signal becomes an investment outcome only after portfolio construction, sizing, execution, and risk management. Those choices are especially consequential in merger arbitrage. The strategy earns a spread in ordinary states but is exposed to discrete losses when transactions fail (Mitchell and Pulvino, 2001); risk-bearing capacity and deal-completion risk are therefore central to both returns and limits to arbitrage (Baker and Savaşoğlu, 2002). A system can read most news correctly and still be

a poor delegate if it sizes and manages a rare mistake badly. That is exactly why training on the range of possible outcomes, and not just on the deals that historically closed, matters.

We build on a rapidly growing literature on machine learning and artificial intelligence in financial markets. Applied to trading, most analyses have compared trading algorithms and their effects on markets, for example, on liquidity (Colliard, Foucault, and Lovo, 2026), and on potential collusion and price efficiency (Dou, Goldstein, and Ji, 2023). However, we are interested in the potential interaction between humans and machines and whether machines can outperform humans, a priority emphasized by Vega, Xiu, and Yang (2026). Dou, Goldstein, Li, and Yang (2026) characterize equilibria analytically in a market for a single risky asset in which reinforcement-learning AI traders compete with humans under alternative information structures. The AI traders' algorithms are outperformed by the most sophisticated humans unless the AI traders hold a private-information advantage.

Our interest is the delegation of the investment strategy itself. In a comprehensive strategic asset allocation (SAA) application, Ang, Azimbayev, and Kim (2026) use AI agents to develop a self-driving portfolio governed by an Investment Policy Statement, a process they refer to as an "agentic strategic asset allocation pipeline". One of their conclusions is that "whether agentic SAA results in better performance than traditional human-centered SAA can only be answered by live performance". Our evaluation on simulator-generated test paths (representing potential out-of-sample paths) that are controlled, replicable, and directly comparable to human decision makers offers a complement to waiting for live performance. In our setting, the agent must form the investment strategy from the case materials and news, validate it against noisy outcomes across many simulated paths, and commit to it. The locked strategy is then judged on the out-of-sample realized profit and on the risk taken to earn it. Because both groups are scored on identical news and price paths, the human benchmark is a realized frontier rather than a modeled one.

In a replayable merger-arbitrage market, we field 45 autonomous agents – built on four frontier language models and replicated across each model's reasoning-effort settings – and evaluate their locked strategies on ten saved realizations from the 2026 Rotman International Trading Competition, each traded by 74 individual human accounts. Amongst other results, three general findings stand out.

First, autonomous strategy formation is technically successful but economically incomplete. Every agent produces a working strategy and every model-effort configuration earns positive mean profit out of sample, yet the average agent beats only 35% of human accounts on the identical realizations, the strongest agent achieves the 60th human percentile, and no agent places in the human top three in any realization. This gap should not be read as a contest between autonomous AI agents versus humans without AI. As noted above, the human competitors act as principals who were free to use any tools, including AI, so the comparison measures how much human guidance, supervision, and governance still add to autonomous agents, not whether machines can match unaided people.

Second, robustness on simulated histories carries over out of sample: rank from practice has a correlation of 0.50 with rank in the test realizations, and the eight best practice performers beat 12.6 percentage points more of the human field than the remaining agents.

Third, replaying archived strategies from each build session on the test paths shows that the strategies improved over the sequence of build sessions, from beating 24% of human accounts to 35%, mainly as agents scaled up their positions after observing the first human benchmarks. The strategies that finished best paired this scaling with targeted restrictions on the specific trades behind earlier losses. Therefore, risk management, not signal reading, was the discriminating skill. Importantly, a model label did not consistently identify those best strategies. Replicate agents built on the same model with the same settings could produce materially different policies and tail risks. Hence the value of human selection of strategies and on-going supervision.

Our paper contributes in several ways. First, and most importantly, it provides a simulation-based methodology for building and then cleanly evaluating decision strategies, whether developed by AI agents or by humans. Training on draws from a calibrated environment, locking the strategy before evaluation, and testing it on realizations that human decision makers also faced prevents data snooping and look-ahead bias by construction rather than correcting for them statistically, and makes agent versus human rankings directly comparable. The price is model risk with respect to the calibrated environment, which we discuss in Section 2.1. The methodology is general: any decision task that can be represented in a calibrated, replayable simulator can be studied the same way.

Second, as in Ang, Azimbayev, and Kim (2026), our paper extends the finance literature on LLM information processing from a forecast or score to an end-to-end delegated investment task. Further, our methodology separates our design from work that invokes the model inside the decision loop which then relies on live performance data to build and evaluate the strategy. In our application, reasoning occurs before execution, including research, modeling, experimentation, testing, and consolidation during strategy formation. Strategies are then evaluated on test realizations representing potential out-of-sample paths.

Third, the paper contributes to performance measurement and manager selection. Prior returns are notoriously noisy measures of human investment skill (Carhart, 1997; Kosowski, Timmermann, Wermers, and White, 2006; Barras, Scaillet, and Wermers, 2010). The same problem applies to AI strategies because repeated model invocations can produce different policies even under a common mandate. Our replayable laboratory supplies multiple shared shocks, holds execution conditions fixed, preserves weak and failed strategies rather than selecting on survival, and then evaluates the locked strategies on unseen paths. This makes performance persistence and selection measurable at the strategy level, with a level of experimental control that real-world trading comparisons cannot achieve.

Fourth, the paper contributes evidence on the governance of autonomous economic agents. Emerging work emphasizes that agents often act on behalf of principals and that their objectives are shaped by instructions, feedback, and institutional constraints (Hadfield and Koh, 2025). Our evidence shows why supervision cannot be reduced to checking whether an agent understood the decision task and relevant text. Benchmark feedback changes the scale of risk; model labels explain only part of cross-agent variation; and results-blind self-assessments are only modestly informative about realized standing. External validation and enforced risk limits remain economically valuable even when the system can write and critique its own strategy.

The remainder of the paper proceeds as follows. Section 2 sets out the methodology, the case, the agent population, the build phase, and the realization-matched human comparison. Section 3 reports out-of-sample performance against the human field and its heterogeneity across model–effort cells and replicate agents. Section 4 studies whether practice performance on simulated histories predicts out-of-sample standing. Section 5 replays the strategies from

the sequence of build intervals on the test (out-of-sample) paths to measure what the agents learned, documents tail risk, and uses the journals to interpret strategy formation. Section 6 evaluates results-blind self-assessment, and Section 7 concludes.

2 Methodology and Research Design

2.1 Two stages: build on many histories, test out of sample

Our design separates the two tasks that a historical backtest combines: developing a strategy and measuring how well it performs. Figure 1 summarizes our protocol. Panel A of Table 1 contrasts it with a historical backtest and Panel B contrasts our comparisons of AI and human performance with those based on market data.

Let ω denote a complete realization (sample path) of the case: the sequence and content of news, the resulting path of each deal's completion probability (including whether and when a deal completes or fails), and the resulting price paths of every security. Throughout, a realization always means an entire sample path ω, never a single draw of one variable. Participants never see a realization in advance; it unfolds over 600 one-second ticks (one iteration of the case; in the competition, a heat), so a trader experiences it as a live market.

The simulator generates realizations from a distribution F designed to reproduce the features of real-world merger arbitrage documented in the literature. Its news categories correspond to the sources of completion risk that drive merger-arbitrage returns (Mitchell and Pulvino, 2001; Baker and Savaşoğlu, 2002). The news items describe the institutions and procedures through which that risk is resolved in practice. Negative news moves a deal's completion probability more than positive news of the same severity, consistent with the large losses on failed deals documented by Mitchell and Pulvino (2001).

The realized historical record corresponds to a single draw, ω^{hist}. A decision strategy (policy) π selected to maximize profit $r(\pi, \omega^{\text{hist}})$ on that draw, and then evaluated on the same draw, can report an in-sample maximum that is biased upward, and the bias grows with the number of candidate policies tried. Such a strategy can also be fragile in ways that the historical path never revealed, because adverse events in the tail of F (a failed financing,

a regulatory block, a competing bid) may simply not have occurred in ω^{hist}.

It is useful to separate backtesting from data snooping. A backtest is a procedure: it computes $r(\pi, \omega^{\text{hist}})$ for a candidate policy. Data snooping can result in a bias that arises when the same draw is used both to search over candidate policies and to judge the one selected (Lo and MacKinlay, 1990; Sullivan, Timmermann, and White, 1999). Backtest overfitting can result from data snooping (Bailey, Borwein, López de Prado, and Zhu, 2014). The finance literature has mainly treated snooping as a problem to be corrected after the fact. Multiple-testing procedures ask whether the best of the policies tried beats a benchmark once their number is taken into account (White, 2000; Hansen, 2005; Romano and Wolf, 2005). Higher significance hurdles (Harvey, Liu, and Zhu, 2016) and the deflated Sharpe ratio (Bailey and López de Prado, 2014) adjust the reported statistic, and hold-out or walk-forward samples reserve part of the record for evaluation.

Each of these remedies takes ω^{hist} as given and therefore shares four limits. First, the corrections require the number of policies tried, which is rarely recorded and may include variants tried by earlier researchers. Second, no correction can supply the tail events that ω^{hist} does not contain. Third, a hold-out segment of a single history can be used only once: after its results inform a revision, it becomes part of the search. It also may not be representative of the distribution of outcomes that could occur in the future. Fourth, for language-model agents, ω^{hist} may lie inside the model's training data, so apparent foresight can be memory (Glasserman and Lin, 2023). No multiple-testing adjustment removes that look-ahead bias. Our design instead replaces the single draw with many draws from F and separates development from evaluation by construction.

Stage 1: building a robust strategy (a Monte Carlo view of history). During the build phase, each participant develops its strategy using a set of realizations

$$\mathcal{D}^{\text{build}} = \{\omega_1, \ldots, \omega_N\}, \qquad \omega_n \sim F, \tag{1}$$

consisting of all the practice runs it chooses to make within the seven build intervals together with the 12 realizations of the six scored practice sessions. Therefore, N varies across par-

ticipants. Each draw shows the participant a different version of what could have happened under the same deal structures and economic mechanics. The objective the participant is effectively asked to optimize is therefore performance across the distribution, approximately $\mathbb{E}_F[\,\cdot\,]$, rather than performance on one path.

Feedback after each scored session, which includes the agent's own P&L together with the human benchmark on the same realizations, shows what level of performance other participants achieved and where the agent's strategy fell short. The next build interval allows the participant to rebuild accordingly. After the seven build intervals, the first six each followed by a scored session, each agent's best strategy π^* is locked.

Stage 2: locking the strategy and evaluating it out of sample. The locked strategy π^* depends, by construction, only on information in $\mathcal{D}^{\text{build}}$. It is then evaluated without revision on ten test realizations,

$$\mathcal{D}^{\text{test}} = \{\omega_1^{\text{test}}, \ldots, \omega_{10}^{\text{test}}\}, \tag{2}$$

that were generated separately from the build phase and never revealed before the lock. Each ω_h^{test} is an out-of-sample scenario: a draw obtained by projecting the calibrated distribution F forward to the period after the lock. Where Stage 1 offers a Monte Carlo view of history, Stage 2 offers a Monte Carlo view of the future, the potential paths a deployed strategy could face.

The out-of-sample score of an AI agent is its average rank across the ten test realizations,

$$\bar{R}(\pi^*) = \frac{1}{10}\sum_{h=1}^{10} \text{rank}\left(r(\pi^*, \omega_h^{\text{test}})\right), \tag{3}$$

where ranks are taken against the 45 agents trading the same realization, and its placement in the human field for each realization is

$$H(\pi^*, \omega_h^{\text{test}}) = \frac{100}{74}\sum_{j=1}^{74} \mathbf{1}\left\{r(\pi^*, \omega_h^{\text{test}}) \geq r_j(\omega_h^{\text{test}})\right\}, \tag{4}$$

the percentage of the 74 individual human accounts trading the identical realization whose

P&L the agent met or exceeded. Because π^* is fixed before any test realization is revealed, these scores contain no look-ahead bias from strategy development, from the agent's training data, or from feedback during evaluation. Because the scores average over ten realizations, they describe performance over a range of outcomes rather than a single draw. And because participants are price takers whose orders do not alter the saved price paths, $r_j(\omega_h^{\text{test}})$ is the actual P&L a human account earned on exactly the realization that the locked AI strategy trades.

Relation to data-snooping corrections. The design therefore prevents data snooping rather than correcting for it. Because π^* is fixed before $\mathcal{D}^{\text{test}}$ is revealed, the validity of $\bar{R}(\pi^*)$ and $H(\pi^*, \omega_h^{\text{test}})$ does not depend on how many variants a participant tried in Stage 1, so no adjustment for the number of trials is needed. The archive nonetheless records every policy (decision strategy) build, so the Stage 1 search is fully observed. Tail events enter both $\mathcal{D}^{\text{build}}$ and $\mathcal{D}^{\text{test}}$ as draws from F rather than depending on what happened historically. And because every realization is generated by the simulator, no model can have seen it during pretraining.

The design does not remove every concern, and three remain. First, it exchanges data-snooping risk for model risk. Performance is out of sample with respect to F, and robustness across draws from F implies robustness in markets to the extent that F is well specified for the particular application. This is why its parameters should be calibrated to real-world evidence so that the draws from F will mimic the relevant real-world distributions. Second, the ten realizations in $\mathcal{D}^{\text{test}}$ are used in every analysis in Sections 3 to 6, so researcher choices made on those test realizations can reintroduce selection. We therefore report selection results at several cutoffs, with a bootstrap over realizations and a cutoff-free rank correlation (Section 4), and we read rankings of individual agents on $\mathcal{D}^{\text{test}}$, such as Table 4, as descriptive. Third, the test realizations are not an entirely new environment, nor are they strictly draws from the build-phase F. They were drawn with a parallel news bank prepared for the competition, which has the same structure, probabilities of occurrence, and price process as the build-phase bank, but with new text describing a different event of the same category for about half of its items. Although half of the distinct news items in the competition

also appear in the build phase, they recur in new combinations, orders, and timings: a test realization typically shares about five of its roughly 50 news items with a given build-phase realization, and never more than 28 percent. Stage 2 therefore tests generalization to new realizations and partly new language, not to a new asset-pricing problem. This is exactly the situation of a delegate deployed after training: the rules of the market are known, the next path is not.

The same logic underlies pilot training. The build phase is the simulator curriculum, in which trainees face many routine and unusual scenarios and are debriefed after each session (analogous to feedback from the scored sessions in our application). Our locked strategy and test realizations are the test flight, flown on scenarios that were not rehearsed during training. The two stages together let us ask not only whether a strategy did well, but whether it was built to do well across the outcomes that could occur, and whether that robustness survives contact with genuinely new information.

2.2 The merger-arbitrage decision problem

Our trading environment is the merger-arbitrage case developed for the Rotman Interactive Trader (RIT) Market Simulator in the configuration used for the RITC 2026 competition, as specified in the RITC 2026 Case Package (Rotman School of Management, 2026a). Mak and McCurdy (2019) describe the simulator and its decision-case methodology more generally. Each realization of the merger-arbitrage case lasts 600 one-second ticks and contains five live merger deals and ten tradable equities: one target and one acquirer for each deal. Deal structures include all-cash, stock-for-stock, and mixed consideration. At the beginning of a path, participants observe deal terms, initial prices, and initial completion-probability estimates. They then receive news: deal-specific text updates about regulation, financing, shareholder actions, competing bids, and process or timing developments, together with market-wide items that move all five deals at once. A path contains about 50 such items, drawn from a bank of 365 and arriving at random times; some are ambiguous in direction, and some can occur only as later steps of a storyline begun by earlier items. The items were researched from public sources on merger processes and drafted with the help of large language models. The stochastic processes generating these news arrivals, their effects on

completion probabilities, and prices are what the simulator draws from in producing each realization.

Economically, the case lies at the intersection of two classic ideas. The no-arbitrage logic of Ross (1976) supplies a benchmark tying relative values to state-contingent payoffs, while Ross (1989) connects the arrival and resolution of information to price variation. Merger arbitrage is not riskless arbitrage: announced deal terms link the target and acquirer payoffs conditional on completion, but news changes the probability of that state and failure produces a discrete break payoff. The case makes this updating problem operational through the following valuation benchmark. For deal d at tick t, the target's benchmark intrinsic value is

$$P^*_{d,t} = p_{d,t}K_{d,t} + (1 - p_{d,t})V_d, \tag{5}$$

where $p_{d,t}$ is the perceived probability of completion, $K_{d,t}$ is the consideration if the deal closes, and V_d is the target's standalone value if the deal fails. For cash offers, $K_{d,t}$ is fixed. For stock and mixed offers, it varies with the acquirer's price. The case brief describes the market structure, the news categories, and a framework for probability updating, including the rule by which news of a given category and severity changes a deal's completion probability. It does not disclose which news items exist, how likely they are, or when they arrive, and it leaves implementation, signal classification, hedging, position management, and allocation across deals to the participant.

The typical approach to modeling merger-arbitrage equilibria is to use news and Bayesian updating about deal completion. Arbitrageurs have prior probabilities about success and iteratively revise them to posterior probabilities in response to noisy signals (news). Market prices then clear at an equilibrium discounted price reflecting the updated probabilities. In this setting, the human participants typically build automated, semi-automated, or discretionary systems to interpret deal news, infer intrinsic values, update completion probabilities, and trade target and acquirer securities under uncertainty.

Each participant's account has a 100,000-share gross position limit, a 50,000-share net limit, a 5,000-share maximum order size, and a commission of two cents per share. Positions and P&L reset at the end of each path. The outcome of a path is end-of-period P&L after

the simulator closes remaining positions at the final tick at the prevailing market price. By design, most deals are still pending at the final close-out (only 3 of the 50 deal-paths in the competition reached completion or failure), so end-of-period P&L marks pending positions at prices that reflect each deal's completion probability. P&L therefore depends on how those probabilities evolve over the path, not on the initial deal terms or on resolved outcomes alone. The case therefore tests the ability to update beliefs, manage risk, and allocate capital in a dynamic environment with incomplete information. Participants are price takers: orders execute against quotes without changing the saved price path. This property makes replay comparisons possible and also means that the 'competition' is a tournament over performance on common shocks rather than a game of endogenous price impact.

2.3 Agents as delegated strategy developers

Our paper is about comparing AI agent and human performance, not about comparing alternative AI model vendors. Table 2 reports the AI population we used, fielding 45 autonomous AI agents from 4 models, which we label AI_1, AI_2, AI_3, and AI_4, associated with 2 providers (Anthropic and OpenAI). Each of the 4 models is crossed with its provider-supported reasoning-effort settings, with three replicate invocations per cell. The effort grids differ across providers and their labels are not assumed to be cardinally comparable: a 'medium' effort setting for one provider may not be directly comparable to a 'medium' effort level in another. Therefore, comparisons across model families capture the bundled differences among entire model systems, including their training, architecture, and tool-use behaviour, rather than the effect of any single design choice. Effort-level comparisons are meaningful only within a family, where the labels are ordered by the same model, for example, AI_1-low, AI_1-medium, etc.

Every agent receives the same common mandate: prepare for and perform as well as possible in the competition. It also receives the same case brief, official scoring methodology, competition-format notice, and API documentation. The scoring document explains that P&L is converted to an ordinal rank for each realization and that the average rank across the realizations determines standing (Rotman School of Management, 2026b). Agents are free to decide how to research, practise, write code, keep notes, and revise. They cannot access

outside information, other agents' files, or researcher-side results. The harness thus gives the AI agents the same information and preparation opportunities that the human participants received, and nothing more.

The AI model acts as a strategy developer rather than as a tick-by-tick trader: between practice runs it reasons and changes code; while a path unfolds, its code observes prices and news and submits orders without another model call. Specifically, the agent writes a Python strategy whose tick handler receives prices, positions, account value, and accumulated news and then returns orders through a fixed interface. After a practice run, the agent receives its final P&L and an agent-readable log of quotes, positions, and news. During build intervals, the agent can inspect its own materials, revise the strategy, run another practice path, and maintain a private learning journal. This structure treats the LLM as a delegated quantitative researcher that develops a policy rather than as a trader reacting inside the market clock. It also makes the lock in Stage 2 exact: the object evaluated out of sample is a fixed program, not a model that could keep changing during evaluation.

Agents operate in private workspaces and under blinded numeric account IDs. Strategy executions occur in locked-down containers with no host file-system access and network egress restricted to the exchange interface. The research record outside an individual agent's reach includes other agents' model calls, tool calls, strategy versions, practice runs, and the mapping from their scored results to the exact code that generated them.

2.4 Stage 1: Building strategies across simulated histories

The protocol (summarized in Figure 1) contains seven build intervals separated by six scored practice sessions, mirroring the preparation schedule the human participants followed ahead of RITC 2026. Within a build interval, each practice run presents the agent with a new path drawn from the case's stochastic environment, so an agent that practises repeatedly confronts a variety of deal outcomes, news sequences, and price paths. At a session boundary, the current strategy is locked and evaluated on two saved realizations previously faced by the human cohort during its own practice. The agent then receives its own P&L outcomes from the two scored practice realizations and the corresponding human top-ten account leaderboards. It does not receive other agents' code or other agents' private reasoning. A

short, fixed reflection window follows, and the next build interval permits further revision. Over the build phase, the agents therefore learn from many simulated histories, and the scored sessions give them a realization-matched view of where they stand relative to human practice performance.

2.5 Stage 2: Locked strategies and the human comparison

After the final build interval, each agent's strategy is locked and evaluated on the ten test realizations of Section 2.1, which are the saved realizations from the RITC 2026 merger-arbitrage competition. Replaying them gives every locked AI strategy exactly the news sequences and price paths that the human competitors faced, and because the market configuration supplies liquidity exogenously and participant orders do not move the price path, an agent's result can be placed directly into the corresponding field of individual human accounts. No competition result or leaderboard is disclosed until all ten realizations have been traded. The deal structures and economic mechanics disclosed in advance are retained in the competition; the extent of any news overlap with the build phase is discussed in Section 2.1.

Outcomes are scored by the average rank $\bar{R}$ and the human placement H defined in equations (3) and (4), treating each agent as a one-member team under the competition's rank-and-average principle. Ranking within a realization removes differences in dollar opportunity scale across realizations, so we report mean P&L per realization alongside the ranks to keep dollar magnitudes visible. The same rank construction, applied to the 12 scored practice-session realizations (six sessions with two realizations per session), serves as the pre-competition signal of Section 4.

The human field comprises the 148 individual trader accounts who traded the merger-arbitrage case at RITC 2026, a global intercollegiate trading competition hosted by the Rotman School of Management. The case ran in two rounds of five heats, which supply our ten test realizations. Each of the 37 participating university teams fielded two trader accounts in each round, so every realization was traded by 74 human accounts. Teams received the case package in advance and practised on the schedule that the AI agents' build intervals replicate, competing for prize money and standing. Team members could coordinate, and official scoring aggregated a team's accounts. Our benchmark instead compares individual

accounts, so that account-level capital and trading constraints match. The term 'realization-matched' is therefore deliberate. Human competitors could collaborate within teams, use external analytical tools (including AI), and prepare outside the agent sandbox; the AI agents faced study-specific budget ceilings and could not run offline tests. The evaluation holds the market state and account-level rules fixed, not the broader production process by which human and machine strategies were created.

The human field is therefore best understood as a measuring stick rather than an opponent or a control group. Its accounts belong to prepared, coordinated teams competing for real stakes, which is what makes the standard demanding; and because the replay leaves the saved price paths undisturbed, each locked agent strategy can be placed into the field after the fact without altering it. What the placement measures is correspondingly precise: where an agent's locked decision strategy stands among the human participants who traded the identical realization.

3 Do AI Agents Reach the Human Frontier?

The test competition supplies the paper's most economically legible test. Every agent strategy is locked before the first test realization, each realization contains news that the agent has not previously observed, and every agent-human comparison is made on the same saved realization. The results in this section are therefore out-of-sample measures of the strategies the agents built, evaluated over ten realizations rather than one. Table 3 reports outcomes for each model-effort cell with three replicate agents per cell. Panel (a) of Figure 2 shows all 450 agent-realization outcomes reported as each agent's percentile placement with respect to the 74 individual human outcomes for the corresponding test realization. Panel (b) averages those outcomes across the ten realizations to show each agent's mean placement in the human field. The next three subsections summarize the results.

3.1 Agent capability–human frontier gap

The agents clear a non-trivial implementation threshold. Every model–effort cell earns positive mean P&L in the competition, from 32.9 thousand dollars per realization for AI_4-low to

337.0 thousand for AI_3-medium, and the average agent earns 134.6 thousand per realization. Nearly every agent trades in most realizations, and the strategies jointly implement news classification, position selection, order submission, and risk rules without a model call during execution. This is stronger evidence of operational capability than accuracy on a static exercise since an error can incur unnecessary commissions, create an unhedged position, or persist until the end of the path.

Operational success does not imply parity with the human field. Across all agent-realizations, the average agent meets or exceeds 35% of human accounts. The best agent, P35, averages 60%. It is above the human median in most, but not all, realizations and never enters the human top ten. This gap is consistent with the analytical result of Dou, Goldstein, Li, and Yang (2026), discussed in Section 1, that AI traders are outperformed by human traders unless the AI agents hold a private-information advantage. In our design, no AI agent has one; and, since our application involves building an entire merger-arbitrage strategy, the potential advantage of humans with access to AI tools could be enhanced relative to just trading an individual security. Across the full population, agents generate 8 human-top-ten placements out of 450 opportunities (six by AI_2 strategies and two by AI_3 strategies) and 0 top-three placements. The occasional top-ten outcomes belong primarily to concentrated strategies on favourable realizations; they do not translate into frontier performance over the complete set of shocks. This is the value of evaluating on a range of realizations: a single realization could have crowned a concentrated strategy that the full set of test realizations exposes as fragile.

The distance is economically large in dollar terms as well. Top human accounts often earn on a scale not reached by the leading agents. A few machine strategies approach that scale by using almost the full gross limit in a single deal, but those same strategies incur deep losses on other paths. The human comparison therefore reveals more than a gap in average performance. The strongest human accounts earn at the scale the concentrated strategies occasionally reach, while avoiding the losses those strategies incur on adverse paths: a frontier defined by the joint management of opportunity and downside that no AI agent attains consistently.

3.2 Why the human benchmark changes the assessment

If the paper reported only absolute profitability, every model family would earn positive mean P&L and therefore appear successful. But a positive result can be misleading on an opportunity-rich path: a strategy may be profitable while still capturing only a small share of the available opportunity.

The human percentile introduced in Section 2 corrects for this by re-scoring each outcome against what participants actually extracted from the same realization. This comparison is available only because the evaluation is realization-matched; with market data, the human and machine records would come from different realizations. Our approach separates two achievements that absolute P&L combines: completing the case, which nearly every agent managed, and exercising competitive investment judgment, which is the more demanding standard applied in this section.

This distinction is central for deployment. A principal does not usually ask whether an automated strategy can place legal orders or produce a positive backtest. Rather, the principal asks whether the strategy adds value relative to available human or systematic alternatives. In this study, the agents demonstrate broad functional competence, while the human field, whose members could combine their own judgment with AI tools, shows that the best attainable policies remain materially stronger.

3.3 Heterogeneity across model–effort cells and replicates

Averages mask substantial cross-agent variation. Because our question is how specific agents compare with the human field, not which vendor's model is best, we report heterogeneity at the level at which agents were fielded, the model-effort cell, and across replicates within a cell. Two cells stand out in Table 3. The average ranks for the AI_3-medium and AI_3-high cells were 12.2 and 14.5, respectively, with human placements of 50.3% and 47.9%. They were the only cells near the human median. In absolute dollar terms they averaged P&L of 337.0 and 279.0 thousand dollars per realization, respectively. Every other cell averaged a rank between 19.8 and 30.8 and beat between 23.0% and 39.1% of human accounts. Note that the model family label does not guarantee these outcomes: AI_3-low averages rank 26.8

and earns 71.7 thousand dollars per realization, behind most cells built on the other models. Nor is high standing confined to one provider: Table 4 shows that the top ten agents include three AI_1 and two AI_2 strategies as well as five AI_3 strategies.

Average cell estimates are imprecise by construction. Each cell contains three replicates and the ten realizations differ in the opportunities they offer. The replicate dispersion is itself economically relevant. Two agents with the same model and effort setting can leave formation (from the build intervals) with different code, different interpretations of feedback, and materially different tail behaviour. The foundation model label therefore does not identify the policy a principal will receive. Independent formation runs and policy-level validation on test realizations will remain valuable even when the deploying institution has already chosen a model vendor; Section 4 asks whether such validation can be done before deployment.

Nor does activity volume explain performance. Under identical time budgets, practice-cycle counts differ sharply across models (Table 2): the AI_3 agents, which include the two leading cells, averaged 82 practice cycles and wrote the shortest strategies and journals in the study, against 180 cycles for AI_1 agents, while AI_4 agents ran the fewest cycles (68) and include the weakest cell. Cycle counts combine model speed, tool-use, voluntary stopping, code-writing time, and research choice. Moreover, within some families, the agents completing more practice cycles perform better. Establishing a deliberation–experimentation tradeoff would require a design that holds the foundation model fixed and randomly assigns separate thinking and market-run budgets. The present data support the narrower conclusion that brute activity volume does not account for out-of-sample standing: what an agent learns from each simulated path matters more than how many paths it runs.

Finally, the role of reasoning effort differs from that in live decision-time systems, where additional reasoning can improve choice but impose a latency cost (Cheng, Granger, Shi, and Strela, 2026). In our application, the LLM is outside the trading clock (reasoning occurs before execution), so reasoning effort affects the process that produces the compiled policy. Relevant deployment risks are policy variation, formation cost, and robustness to new paths. Future comparisons could report not only performance but also implementation costs, such as elapsed formation time and API usage costs.

4 Does Practice Rank Predict Out-of-Sample Rank?

Delegating capital requires a selection rule. Model identity is one candidate, but a natural alternative is to run candidate policies on common development shocks and select those that perform well. This is the machine-strategy analogue of using a manager's track record, with the same concern that observed returns may reflect luck rather than skill (Carhart, 1997; Kosowski, Timmermann, Wermers, and White, 2006; Barras, Scaillet, and Wermers, 2010). Our two-stage design makes the question clean. The selection signal is computed entirely from Stage 1, on simulated histories, and the outcome is measured entirely in Stage 2, on test realizations that no strategy had seen; there is no channel through which the outcome can contaminate the signal.

Our pre-competition score ranks every agent within each of the 12 scored practice-session realizations and averages those ranks. It uses only information available before the final strategy is locked. The outcome score (average rank) applies the same procedure to the ten competition realizations, which were run with strategies locked and whose news sequences were not available during formation.

Figure 3 plots the two scores (average ranks) for all four model families. Their rank correlation is 0.50. The association is economically visible rather than being driven solely by the final winner: most of the leading AI_3 strategies lie in the favourable part of both rankings, and several weak practice performers remain weak in competition. At the same time, there are consequential exceptions. Two members of the illustrative pre-competition top eight finish in the middle of the final agent field, while some agents outside that set improve their relative standing. Selection reduces but does not remove policy risk.

Table 5 reports illustrative top-k rules. The best eight by pre-competition rank average 15.8 in the competition, compared with 24.6 for the remainder. They beat 45.4% of human accounts, compared with 32.8% for the remainder. The relation is not specific to eight: the table reports five, ten, and fifteen as well.

The selection result is useful for three reasons. First, all candidates face the same 12 development realizations, so relative performance is not confounded by different market shocks. Second, the final evaluation includes partially new news language. Persistence therefore re-

quires more than memorizing a practice headline; it requires a policy that continues to map new text and prices into profitable positions, which is what training across many simulated paths is meant to produce. Third, failed candidates remain in the sample. There is no survivorship screen in which only deployed winners become observable.

The selected agent is the entire generated policy (model, effort setting, replicate history, code, and risk rules) rather than the foundation model alone. Note that ten test realizations will not fully reveal a strategy's full return distribution. Because the simulator can generate further realizations from the same calibrated environment, a larger replay bank would make the selection exercise closer to a conventional out-of-sample portfolio sort and permit direct estimation of false-selection rates. However, even in a noisy environment, observed common-shock performance contains more useful information than a model label alone. An institution delegating capital to autonomous strategies can therefore evaluate them the way it evaluates candidate managers, or the way an airline certifies pilots: train several independent instances on common simulated paths, score them on matched shocks, retain the complete performance distribution, lock the candidates, and select policies on out-of-sample performance rather than assuming that one invocation represents a model family.

Section 5.1 shows that practice also improved each agent's own strategy, mainly through scale.

5 Competitive Benchmarks and Risk Analysis

The longitudinal record of the build phase allows us to look inside the policy-formation process. We first measure what changed in the strategies, by replaying every archived strategy from each build interval on the test realizations, then examine how the leading strategies manage the risk that came with those changes, and finally use agent journals to interpret why policies changed. This ordering matters: realized P&L on a new path is not itself evidence that a risk control was added or removed.

5.1 Benchmark feedback, exposure, and what the agents learned

Session 1 establishes that most agents can generate positive P&L but operate on a smaller scale than the visible human leaders. The agents built sensible, working merger-arbitrage strategies from the case brief alone, but until the human leaderboard made the competitive frontier concrete, the goal of winning was not automatically embedded in those strategies. After receiving their own two results and the two human top-ten lists, agents are allowed to reflect and revise. Table 6 records what they then did on the practice paths they actually traded, using sampled maximum gross exposure and sampled within-realization drawdown computed from the archived execution record. These measures reflect what each strategy actually did rather than anyone's reading of its code or journal. Between Sessions 1 and 2, mean maximum gross exposure rises from 22.1 to 35.2 thousand shares, and 36 of the 45 agents (80%) increase exposure (paired Wilcoxon signed-rank test, $p < 0.001$); the step from Session 2 to Session 3 is at least as large in absolute terms (84% of agents increase exposure), and an agent's Session 1–2 increase is unrelated to its Session 1 P&L (rank correlation -0.03). Practice P&L cannot show whether these changes improved the strategies, however. Every scored session presents new paths, so cross-session P&L combines policy change with opportunity: mean practice P&L more than doubles from Session 1 to Session 2 and then falls in Session 3, although exposure keeps rising.

Our test realizations remove that problem. Because they are saved and participants' orders do not move prices, any archived strategy can be traded on them after the fact. We therefore replay, on the same ten test realizations, every agent's strategy as scored in each of the six practice sessions (versions S1 to S6) together with its locked competition strategy (Final): seven versions for each of the 45 agents, or 3,150 agent-realizations. Within an agent, the versions differ only in code, so any difference in outcome is a difference in the strategy. Each outcome is placed among the 74 human accounts that traded the same realization, using the human placement H of Section 2.1. The human field is held fixed, so it serves purely as a measuring stick. We do not compare ranks among the agents across versions, because every agent's version changes at once. The measures and tests were specified before the replay was run.

Figure 4 and Table 7 report the results. On identical realizations, later strategies beat more of the human field. Mean placement rises from 24.4% of human accounts for the S1 versions to 35.2% for the Final versions, a gain of 10.8 percentage points (95% bootstrap interval [5.6, 16.5]). The gain is broad: 41 of the 45 agents improve (panel b), every model family gains between 8.9 and 13.1 points, and no agent's best version is its first. It is also concentrated early. The steps that follow the first two human benchmarks, from S1 to S2 and from S2 to S3, add 3.3 and 4.2 points and together account for about 70% of the total; none of the later steps is distinguishable from zero. The families differ in timing (panel a): AI_3 leads from its first version, while AI_4, which remains conservative for much of development, gains mostly after S3. Nor is the gain confined to news the agents had already seen: it is 9.8 points on the five realizations with the smallest share of news items that also appeared in practice, and 11.9 points on the other five. A common set of realizations is what makes this comparison of practice vintages possible, and it supports a direct, if descriptive, claim that the agents' strategies improved over the build phase.

Panel (c) shows what changed. From S1 to Final, mean P&L per realization rises 3.5-fold, while the maximum gross position rises 2.9-fold and the maximum drawdown 3.7-fold, and the largest increases in exposure come in the same two early steps as the gains in placement. Across agents, the gain in placement tracks the increase in exposure (rank correlation 0.64), and the five agents whose maximum gross position changed by less than 10,000 shares gain about one point on average. Profit per unit of exposure rises only modestly, from 1.8 to 2.2 thousand dollars per thousand shares of maximum gross position, and profit per dollar of drawdown does not rise (1.60 for S1 and 1.53 for Final). The share of agent-realizations that end in a loss is unchanged at about 23%, but losses, when they occur, are larger: the lowest decile of outcomes moves from −\$11 thousand to −\$53 thousand. What the agents learned over the build phase is therefore mainly to participate at the scale of the human field without losing more often, which is the gap their journals diagnose, rather than to earn more per unit of risk.

The replay shows what changed in the strategies, not why. Each step bundles the benchmark feedback with additional build time, and there is no comparison group without feedback, so the gains cannot be attributed to the human benchmark. That they are concentrated

in the first two steps, where exposure also rose most, is consistent with the benchmark mattering but does not establish it; the journals, discussed below, are the evidence that connects the two. The ten test realizations are reused here, which is why the measures were fixed in advance and the results are read as descriptive. Finally, the earlier versions are measured against accounts that traded in the competition, not against the human participants' own strategies at the same stage of preparation, so the placements describe the agents' progress toward the final human standard rather than a contemporaneous contest.

5.2 Risk control heterogeneity in the strategies

The replay shows that the agents learned mainly to trade at scale. What then separates the strategies is how they manage the tail that comes with that scale. Figure 5 relates each locked strategy's competition P&L to its sampled maximum drawdown; the size of the dot represents gross exposure. The leading strategies carry substantial gross positions. Their advantage is that exposure more often produces upside without a commensurate expansion of the loss tail. The extremes illustrate the trade-off the leaders escape. At one pole, the most conservative strategies rarely trade at all (one agent submitted no orders in three of the ten realizations), preserving capital but ranking poorly in a field where most participants profit. At the other, the most concentrated strategies stake nearly the full gross limit on a single deal, finishing first in individual realizations when they were favourable while suffering the study's deepest losses on adverse ones.[1] Evaluated on any single favourable realization, such a strategy would look like a winner; evaluated across the range of test realizations, its tail is visible.

This pattern gives risk management a precise meaning in the study. It is not the minimization of variance or the addition of every possible stop-loss rule. Merger news can cause a discontinuous price gap, so a reactive stop may execute only after the economic loss has occurred. Conversely, a broad stop can force exit from a temporarily losing deal that subsequently closes. Effective controls must restrict particular exposures before a gap

[1]The zero-order agent is P47 (AI_4-high); P28 (AI_2-max) and P43 (AI_4-medium) each submitted no orders in one realization. The concentrated pole comprises P22 (AI_2-high) and P29 and P30 (AI_2-max), which account for the five deepest single-realization losses in the competition; P29 nevertheless finished first in four realizations, each a favourable one.

or distinguish informative termination news from ambiguous process language. The high-performing agents combine participation with targeted ex-ante restrictions; the unsuccessful conservative agents omit too much opportunity, and the unsuccessful lottery agents accept too much unbounded downside.

5.3 What the journals add

The private journals are useful because they record the agents' own interpretations before final outcomes are disclosed. They show broad agreement on the initial competitive message. One leading agent re-reads the scoring rule and writes that "being merely positive is not enough" when humans take large positions.[2] Another describes the gap to human P&L as an "under-participation" problem. The exposure changes above show that these statements were translated into code and positions.

The journals also reveal divergent readings of ordinal scoring; each agent's operative reading at the competition is coded in Table 8. A consistency reading emphasizes that average rank punishes blowups. A lottery reading notes that, within a realization, a million-dollar loss and a ten-thousand-dollar loss can both receive last place and therefore treats marginal downside beyond last place as cheap. A third, resolved, formation path attempts to settle the tension by scaling exposure while forbidding the specific trades responsible for prior losses, losses the agents had encountered on earlier simulated paths. Examples include restrictions on short positions in cash-deal targets when there is no explicit termination news, and drawdown locks defined relative to the agent's own running profit peak. The three readings order the test outcomes: agents coded as resolved average competition rank 16.0 over the ten realizations, against 22.5 for the lottery camp and 26.7 for the consistency camp (Figure 6).

The lottery logic affects individual strategies, but it should not be overgeneralized. Average ordinal competition rank and rank by cumulative P&L have a Spearman correlation of 0.97. Thus the scoring rule does not reorder the entire population. It changes the relative standing of a smaller set of highly variable policies and, more importantly, creates incentives

[2] P35 (AI_3-medium), the eventual leader of the agent standings; the "under-participation" diagnosis that follows is from P44 (AI_4-medium).

that lead some agents to relax their own risk protections.

Agents are blinded from one another's work throughout formation. The taxonomy of Table 8, however, was assembled ex post, after competition outcomes existed and with performance context in the coding record. We therefore use the journals and the coded camps to interpret documented behaviour, not as an outcome-blind statistical predictor; the statistical weight rests on the replay and on the exposure and drawdown records above.

6 Can Agents Assess Their Own Readiness?

External performance measurement may be unnecessary if autonomous agents can reliably identify the quality and risks of their own policies. We evaluate this possibility using a results-blind debrief. After all ten competition realizations have been traded, but before any result is disclosed, every agent receives its own cumulative learning journal and is asked to predict its finishing percentile, probability of finishing first in at least one realization, and competitiveness with the strongest human participants. Tools and market access are closed, and the response cannot affect scoring.

The percentile question did not name a reference field, so we score it against realized standing among the 45 AI agents, the only field in which every agent has a defined finishing position; free-text answers were converted to percentile bands by a language model. Figure 7 compares each agent's predicted band midpoint with its realized percentile. The cross-agent rank correlation between the two is 0.30. Agents therefore possess some information about their relative quality, but the signal is weak. The mean absolute error is 26 percentile points, the average stated band is 19 points wide, and only 16% of bands contain the realized finish.

Aggregate calibration can be misleading. The mean realized percentile across the 45-agent field is mechanically 50, so the mean prediction of 54 is not by itself evidence that agents know their own rank. Individual errors are large and run in both directions, so they largely offset in the average while leaving individual predictions far off: in Figure 7, some agents that predicted top-quarter finishes realized bottom-quarter ones, while others that predicted below-median finishes placed in the top quarter.

Agents were also cautious about the human field: their average stated confidence of

being competitive with the strongest human participants was 24 out of 100. That confidence has rank correlation 0.35 with the percentage of human accounts actually beaten. The predicted probability of finishing first in at least one realization has correlation -0.01 with the realized win indicator. The agents can often identify plausible failure modes in prose, but that qualitative awareness does not translate into well-calibrated probabilities.

For a principal, the comparison between Sections 4 and 6 is instructive. A common-shock performance tournament provides a materially useful selection signal. Self-assessment adds some information but is too noisy to substitute for external evaluation; a pilot's confidence is not a substitute for a test flight. Requiring an agent to articulate uncertainty remains valuable for audit and diagnosis; capital allocation should nevertheless depend on realized out-of-sample validation and enforced limits rather than confidence language alone.

7 Conclusion

We study autonomous AI systems as delegated investment-strategy developers, and we do so with a methodology developed from simulation-based learning and analogous to the way pilots are trained. Before pilots fly a plane full of people, they practise in simulators that confront them with the range of routine and unusual situations they may face, including rare cascading emergencies, and they are then checked on scenarios they have not rehearsed. Our design does the same for investment strategies. In the build phase, agents and human competitors develop their strategies on paths drawn from a simulator calibrated to real-world merger arbitrage, a Monte Carlo view of history that rewards strategies robust to the many paths that could have occurred rather than strategies fitted to the one path that did. Each agent's strategy is then locked and evaluated out of sample, without revision or feedback, on ten test realizations containing new information. Because the strategy is fixed before those realizations are revealed, the evaluation carries no look-ahead bias from model development or training; because it spans ten realizations, it reflects a range of outcomes; and because the saved realizations are exactly those the human competitors traded, every performance ranking compares AI and human decision makers on identical news, price paths, and account constraints. This combination removes the principal obstacles to evaluating machine

and human trading outcomes with market data: different information, confounding market conditions, heterogeneous mandates and constraints, hidden implementation, overlapping training data, the luck of a single path, and the data snooping that arises when that path is used both to build and to judge a strategy.

The evidence from that evaluation draws a line between functional capability and competitive quality. The autonomous AI systems can build and execute economically coherent event-driven strategies. Yet the average agent remains below the human median and even the best agent remains far from the strongest human accounts. Because the human competitors acted as principals free to use any tools, including AI, this gap is not a verdict on humans versus machines. It measures how much human guidance, supervision, and governance can still contribute to autonomous AI agents, which is the central practical lesson for a principal considering delegation. The gap is not well described as a failure to recognize financial language. It emerges from the complete policy: which opportunities the agent enters, how much exposure it takes, and what losses it permits when a rare signal or deal path is adverse. These are exactly the skills that practice across the full distribution of outcomes is meant to build, and the locked out-of-sample test shows how far the agents have, and have not, acquired them.

Performance measurement has value in this setting. Ranks on common practice realizations predict ranks on the competition realizations, and simple top-k rules select strategies that subsequently beat more of the human field. At the same time, replicate variation within the same model–effort cell is large and results-blind self-assessments are poorly resolved. The economically relevant unit of evaluation is therefore the formed and locked policy, not the model name or the agent's description of its own readiness.

Competitive benchmarking also changes the policy-formation problem. After the first human top-ten feedback, agents broadly increase gross exposure, and replaying every archived version on the test paths shows that their strategies improved, from beating 24% of human accounts to 35%, mostly in the first two steps and mainly through scale rather than through more profit per unit of risk. Later sessions separate strategies that remain conservative, strategies that accept lottery-like losses, and strategies that combine large positions with targeted tail restrictions. The paper does not identify the feedback effect causally, but

the replay, the exposure path, and the contemporaneous journals show that risk scale and reference-class inference are central to how autonomous delegates respond to performance pressure.

Inevitably, our application has some limits. There is one case type (our Merger-Arbitrage Decision Case), one competition year (although featuring very well prepared and competitive global participants), ten test realizations from the RITC 2026 final case, which reproduce the complete state of the stochastic environment, and three replicates per model–effort cell for the set of AI agent strategies. Human accounts share the same evaluated states and constraints. We matched the preparation schedule and time allowed but could not fully match the agents' preparation environment; humans can coordinate within teams and use tools, including AI, outside the sandbox. Note that although AI agents were able to benchmark their strategies against the human participants' practice performance, the human participants were not able to observe or adapt to the AI agents' strategies. An extension will allow both to prepare and compete simultaneously. Finally, model families bundle provider-specific differences, and the models themselves will be superseded over time. These features bound claims about a general human–AI ranking or a stable vendor hierarchy, but not the methodology, which can be applied unchanged to new models, new cases, and larger banks of test realizations. More fundamentally, the design exchanges data-snooping risk for model risk: its results are out of sample with respect to the calibrated simulator, and their relevance to live markets depends on how well that calibration captures the real environment.

The methodology also makes several extensions straightforward. The version replay of Section 5.1 is one: because the test realizations are saved, every archived strategy can be re-evaluated on them after the fact. First, one could vary the feedback benchmark (for example, none, human top ten, or full human distribution) while holding model, realizations, and time fixed, to estimate how relative-performance information shapes exposure and tail controls. Combined with the version replay, this would identify how much of the agents' improvement is due to the benchmark itself. Second, because further out-of-sample realizations can be generated by projecting the calibrated distribution forward, the test bank can be enlarged to estimate each locked strategy's full return distribution, including its tail, and to measure false-selection rates directly.

The broader implication is that autonomous financial AI creates a governance problem even when its component capabilities are strong, and that simulation offers a practical way to manage it. An AI agent's autonomous strategy, while technically competent, can be economically incomplete. A principal must specify an objective, choose the feedback distribution, train and validate multiple independent policies on common simulated paths, lock them before evaluating them on unseen ones, and enforce risk constraints that do not depend on the agent's own confidence. Aviation does not certify pilots on the basis of one uneventful flight; investment delegation to AI agents should not be certified on the basis of one historical backtest. The ability to read news is a necessary input. The ability to translate that information into a robust allocation of capital, across the range of things that could happen, is the economically decisive output.

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Table 1: Historical backtests and market-data comparisons versus this design

	Historical backtest or market data	This design
Panel A: Developing and evaluating a strategy		
Development data	One realized path, ω^{hist}	$\mathcal{D}^{\text{build}} = \{\omega_1, \ldots, \omega_N\}$, $\omega_n \sim F$ (Stage 1)
Evaluation data	The same path, or a hold-out segment of it	$\mathcal{D}^{\text{test}}$, test realizations projected from F and generated separately (Stage 2)
Data snooping	Corrected after the fact; the correction requires the number of variants tried	Prevented by construction: π^* locked before $\mathcal{D}^{\text{test}}$ is revealed
Tail events	Only those that occurred on ω^{hist}	Drawn from F in both stages
Look-ahead bias	Evaluation period may overlap strategy development or LLM training data	Simulator-generated paths; new news items, or familiar items in new combinations
Main residual risk	Data snooping and the luck of one path	Model risk: results are out of sample with respect to F, hence its calibration to real-world evidence
Panel B: Comparing AI agents and humans		
Information	Different news sets, timing, and data access	Same case package, news sequences, and price paths
Market conditions	Different periods and opportunity sets	Identical saved realizations for agents and humans
Mandate and incentives	Heterogeneous objectives and benchmarks	Common mandate and ordinal scoring rule; humans compete for prizes, agents directed to win and benchmarked to humans
Capital and constraints	Different capital, limits, and costs	Identical account limits, order size, and commissions
Preparation	Unobserved	Same schedule of build intervals and scored practice sessions (humans may also coordinate and use outside tools)
Price impact	Participants' trades move prices	Price takers; saved price paths unaffected by orders, so results are replayable
Implementation	Hidden	Every model call, strategy version, run, and result archived
Survivorship	Only surviving strategies observed	All 45 agents, including weak and failed strategies, retained

Notes: Panel A contrasts strategy development and evaluation on the single historical path, ω^{hist}, with the two-stage simulator design of Section 2.1, using its notation: F is the calibrated distribution of realizations, $\mathcal{D}^{\text{build}}$ and $\mathcal{D}^{\text{test}}$ are the build and out-of-sample (test) sets, and π^* is the locked strategy. Panel B lists the confounds that make comparisons of AI agent and human investment performance difficult when based on market data, and how the design addresses each. The comparison holds the market state and account-level rules fixed; it does not hold fixed the broader production process by which human teams formed their strategies.

Table 2: Agent population and experimental cells

Model family	Reasoning-effort levels	Cells	Agents	Formation activity: Practice cycles	Formation activity: Code (KB)	Formation activity: Journal (words)
AI_1	low, medium, high, xhigh, max	5	15	180	21.2	4,336
AI_2	low, medium, high, max	4	12	122	24.3	5,507
AI_3	low, medium, high	3	9	82	14.3	898
AI_4	low, medium, high	3	9	68	26.8	1,463
Total		15	45	123	21.8	3,386

Notes: The population spans four frontier model systems crossed with each model's provider-supported reasoning-effort settings. Each cell contains three separate replicate invocations with private persistent workspaces and accounts. Replicate labels are not provider random seeds. All agents receive the same objective, documents, tool interface, and interval-level activity-accounting rule. Provider model identifiers and calls are dated June–July 2026. Formation activity is measured per agent and averaged within family (the Total row averages all 45 agents): Practice cycles counts the practice runs the agent chose to make during the seven build intervals, excluding the scored sessions and competition realizations; Code is the size of the Python source of its final competition strategy, in kilobytes; Journal is the word count of its cumulative learning journal at the end of the study.

Table 3: Out-of-sample competition outcomes by model–effort cell

Family	Effort	N	Mean rank	Rank s.d.	Mean P&L ($k)	Human pct.
AI_1	low	3	25.0	7.1	83.7	31.5
	medium	3	22.6	7.2	132.1	34.7
	high	3	23.6	5.6	105.5	34.1
	xhigh	3	19.8	4.4	146.8	39.1
	max	3	26.3	0.8	83.4	30.0
AI_2	low	3	24.0	0.4	94.2	34.1
	medium	3	25.9	1.2	85.6	32.4
	high	3	24.2	5.3	132.2	34.6
	max	3	22.7	10.2	198.8	38.4
AI_3	low	3	26.8	2.1	71.7	28.8
	medium	3	12.2	3.8	337.0	50.3
	high	3	14.5	6.5	279.0	47.9
AI_4	low	3	30.8	2.8	32.9	23.0
	medium	3	22.6	3.8	130.0	34.0
	high	3	24.1	7.8	106.7	32.3

Notes: Each cell contains the three replicate agents run with the same model and reasoning-effort setting, and every column uses only the ten competition realizations. Each is a saved realization previously faced by the human RITC field, and every agent trades all ten with its strategy locked. Within each realization, the 45 AI agents are ranked by end-of-period P&L from 1 (best) to 45 (worst), with tied P&L sharing the better rank; ranking within a realization controls for the large differences in profit opportunity across realizations. Mean rank averages the three agents' ten-realization average ranks, so lower is better. Rank s.d. is the sample standard deviation of those three agent-level averages. Mean P&L is end-of-period P&L averaged over the cell's 30 agent-realizations, in thousands of dollars. Human pct. is the percentage of the 74 individual human accounts trading the identical realization whose P&L the agent met or exceeded, averaged over the cell's 30 agent-realizations, so higher is better and 50 marks the human median. Effort labels are model-native and comparable only within a family, and family comparisons bundle all differences between the model systems.

Table 4: Top ten agents in the competition

Agent	Family	Effort	Avg. rank	Mean P&L ($k)	Human pct.	Drawdown ($k)
P35	AI_3	medium	7.8	531.5	59.9	97.2
P39	AI_3	high	9.1	364.6	56.9	100.7
P37	AI_3	high	12.6	326.6	50.9	117.3
P36	AI_3	medium	14.0	230.7	45.9	77.5
P04	AI_1	medium	14.4	256.0	46.8	122.1
P34	AI_3	medium	14.7	248.6	45.0	82.8
P10	AI_1	xhigh	15.9	199.6	45.4	58.5
P29	AI_2	max	16.0	302.3	49.5	288.8
P02	AI_1	low	17.3	140.3	41.5	76.9
P30	AI_2	max	17.6	270.4	47.2	248.3

Notes: Agents are ordered by average ordinal rank over the ten competition realizations, run with locked strategies. Human pct. is the mean percentage of individual human accounts met or exceeded on the same realizations. Drawdown is the mean across realizations of the maximum decline from a prior sampled within-run net-liquidation-value peak.

Table 5: Pre-competition selection on simulated histories and competition performance

	Average rank			Human percentile			
Selected k	Selected	Other	Gap	Selected	Other	Gap	95% interval
5	17.0	23.7	−6.7	43.9	33.9	10.0	[2.5, 17.1]
8	15.8	24.6	−8.8	45.4	32.8	12.6	[5.9, 19.4]
10	16.6	24.8	−8.3	44.3	32.4	12.0	[4.9, 18.6]
15	19.3	24.9	−5.6	40.2	32.5	7.7	[1.6, 14.0]

Notes: This table asks whether an investor could have identified the stronger agents before the competition using only the practice performance record from the build phase. In each of the 12 scored practice-session realizations (six sessions of two), the 45 AI agents are ranked by end-of-period P&L from 1 (highest) to 45, with tied P&L sharing the better rank; an agent's pre-competition score is the mean of its 12 ranks. Each row selects the k agents (column 1) with the best average practice ranks across the 12 realizations, breaking ties at the cutoff by higher mean practice P&L. All other columns use only the ten competition realizations, run after all strategies had been locked. Under Average rank, Selected and Other average the within-realization ranks (1 best, 45 worst) over all agent-realizations in each group, and Gap is Selected minus Other, so negative values favour the selected group. For each agent-realization, the human percentile is the percentage of the 74 individual human accounts trading the identical realization whose P&L the agent met or exceeded. Under Human percentile, Selected and Other average that percentile over the same agent-realizations, and Gap is their difference. At $k = 8$, the 80 selected agent-realizations average 45.4, while the 370 remaining agent-realizations average 32.8, a gap of 12.6 points, or about nine additional human accounts beaten per realization (12.6 percent of 74). The displayed interval is a 95% bootstrap for the Gap that resamples the ten realizations with the selected group held fixed; the intervals lie above zero at every cutoff, indicating that the advantage is not driven by a few favourable realizations. The cutoff-free evidence reports the 0.50 rank correlation between the full pre-competition and competition rankings across all 45 agents (agent-bootstrap 95% interval [0.21, 0.72]).

Table 6: Trading intensity and realized risk across scored practice sessions

Session	Mean P&L ($k)	Max gross (000s)	Max drawdown ($k)	Orders
S1	40.1	22.1	27.3	84.3
S2	102.5	35.2	45.4	113.1
S3	63.4	51.2	68.7	161.2
S4	47.2	56.1	51.2	150.0
S5	40.6	64.2	69.0	164.5
S6	72.2	65.2	130.6	134.0

Notes: Each session contains two saved realizations traded by all 45 AI agents, so every row averages over 90 agent-realizations. Mean P&L averages end-of-period P&L, in thousands of dollars. Max gross is the largest sum of absolute share positions observed in the approximately ten-tick sampled account frames of a realization; Max drawdown is the largest decline from a prior sampled within-realization net-liquidation-value peak. Both are conservative sampled measures rather than exact tick-level maxima. Orders is the number of orders per realization that the exchange accepted; the roughly 0.2 percent of attempted orders it rejected are excluded. Every strategy submitted market orders, which execute immediately against the prevailing quote, so accepted orders correspond to executed trades. Strategies may change between sessions after agents observe their own P&L and the human top-ten benchmark. Because realizations also change, the table documents the joint evolution of policies and realized opportunity; Table 7 removes this confound by replaying every version on common paths.

Table 7: Replayed strategy versions: placement, profit, and risk

Panel A: Levels by version

Version	Human pct.	95% interval	Mean P&L ($k)	Max gross (000s)	Max drawdown ($k)	Losing (%)
S1	24.4	[20.4, 28.5]	38.8	21.1	24.3	23.6
S2	27.7	[23.3, 32.2]	61.3	31.6	33.1	18.7
S3	31.9	[26.8, 37.2]	95.0	51.7	67.6	23.1
S4	33.0	[27.8, 38.3]	103.7	54.3	71.7	22.9
S5	33.6	[28.0, 39.7]	112.3	62.2	86.9	24.7
S6	34.9	[29.2, 41.1]	126.3	60.9	86.3	22.9
Final	35.2	[29.3, 41.5]	135.9	61.6	89.1	22.7

Panel B: Change in human placement

Step	Mean change	95% interval	Agents improving	Wilcoxon p
S1 to S2	3.3	[1.7, 5.1]	35	< 0.001
S2 to S3	4.2	[1.5, 7.2]	36	< 0.001
S3 to S4	1.1	[−1.3, 3.7]	22	0.355
S4 to S5	0.6	[−1.8, 2.8]	30	0.077
S5 to S6	1.3	[−1.1, 4.0]	24	0.299
S6 to Final	0.3	[−2.0, 2.5]	27	0.232
S1 to Final	10.8	[5.6, 16.5]	41	< 0.001

Notes: Every row uses the same ten competition realizations. Version Sk is each agent's strategy as locked and scored in practice Session k; Final is its locked competition strategy. Panel A averages over the 450 agent-realizations of each version. Human pct. is the percentage of the 74 individual human accounts trading the identical realization whose P&L the agent met or exceeded; the 95% interval is a bootstrap that resamples agents and realizations independently (20,000 draws). Mean P&L is end-of-period P&L, in thousands of dollars. Max gross and Max drawdown are the sampled measures defined in Table 6. Losing is the percentage of agent-realizations with negative P&L. Panel B reports changes in each agent's ten-realization mean placement: the mean change across the 45 agents, its bootstrap 95% interval, the number of agents whose placement rose, and the p-value of a paired Wilcoxon signed-rank test.

Table 8: Strategy formation: a journal-coded taxonomy

Agent	Effort	Final archetype	Scoring reading	Risk controls	Avg. rank
AI_1					
P01	low	News	Consistency	Added	26.3
P02	low	News	Resolved	Surgical	17.3
P03	low	FV/news	Consistency	Added	31.4
P04	medium	FV/news	Resolved	Surgical	14.4
P05	medium	News	Consistency	Oscillated	27.9
P06	medium	News	Consistency	Oscillated	25.6
P07	high	FV/news	Consistency	Added	30.1
P08	high	News	Consistency	Surgical	20.1
P09	high	News	Consistency	Surgical	20.6
P10	xhigh	FV/news	Resolved	Surgical	15.9
P11	xhigh	News	Consistency	Oscillated	24.5
P12	xhigh	FV/news	Resolved	Surgical	18.9
P13	max	FV/news	Lottery	Oscillated	26.4
P14	max	FV/news	Consistency	Oscillated	25.5
P15	max	Momentum	Lottery	Oscillated	27.0
AI_2					
P16	low	Hold	Lottery	Stripped	24.4
P17	low	Hold	Lottery	Stripped	23.6
P18	low	Hold	Lottery	Surgical	24.1
P19	medium	Hold	Consistency	Stripped	25.5
P20	medium	Hold	Consistency	Stripped	27.3
P21	medium	Hold	Consistency	Stripped	24.9
P22	high	Pair	Lottery	Stripped	18.5
P23	high	FV/news	Consistency	Added	29.0
P24	high	News	Consistency	Added	25.0
P28	max	News	Consistency	Oscillated	34.4
P29	max	Pair	Lottery	Stripped	16.0
P30	max	Hold	Lottery	Stripped	17.6
AI_3					
P31	low	Basket	Consistency	Oscillated	29.1
P32	low	News	Consistency	Added	26.2
P33	low	News	Lottery	Stripped	25.1
P34	medium	Selective	Consistency	Surgical	14.7
P35	medium	FV/news	Resolved	Surgical	7.8
P36	medium	FV/news	Resolved	Oscillated	14.0
P37	high	FV/news	Resolved	Surgical	12.6
P38	high	FV/news	Resolved	Oscillated	21.7
P39	high	FV/news	Resolved	Surgical	9.1
AI_4					
P40	low	Selective	Consistency	Surgical	27.7
P41	low	Selective	Consistency	Added	33.1
P42	low	News	Consistency	Added	31.5
P43	medium	FV/news	Resolved	Surgical	21.4
P44	medium	FV/news	Resolved	Surgical	19.6
P45	medium	News	Consistency	Oscillated	26.9
P46	high	FV/news	Consistency	Surgical	19.6
P47	high	Selective	Consistency	Added	33.2
P48	high	News	Resolved	Surgical	19.5

Notes: This table classifies each agent's strategy formation from its cumulative learning journal. Every journal was read in full and coded against a common rubric whose dimensions include the final strategy design, the agent's reading of the per-realization ordinal scoring rule, and the evolution of its risk controls. Coding was a single retrospective pass in four batches, one per model family, performed by AI-assisted readers working to the authors' rubric. The readers could see each agent's average rank over the first eight competition realizations, so these labels are interpretive rather than outcome-blind; no agent ever saw a competition result. Statistical claims in the text rest on the exposure and drawdown records in Tables 6 and 7, not on these labels. Final archetype is the dominant design of the final competition strategy: FV/news = news signals gated by a fair-value or market-implied-probability model; News = keyword news-sentiment following; Hold = maximum positions taken at the first tick and held to the automatic close-out, exiting only on explicit termination news; Pair = the entire account limit concentrated in one deal's target and acquirer; Selective = trades only rare high-clarity catalysts, otherwise flat; Basket = passive long basket without news signals; Momentum = price-trend following. Scoring reading is the agent's interpretation of per-realization ordinal scoring at its final strategy lock: Consistency = rank-averaging punishes blowups, so stay small and steady; Lottery = last place in a realization costs the same at any loss size, so downside variance is cheap; Resolved = scale up while engineering out the specific loss tail. Risk controls is the net trajectory of stop-losses, position caps and exit rules over the study: Added, Stripped, Surgical (targeted cuts of specific loss mechanisms while keeping the structure) or Oscillated. Avg. rank is the average within-realization rank over all ten competition realizations, 1 being best of 45. Mean Avg. rank is 16.0 for Resolved, 22.5 for Lottery, and 26.7 for Consistency. The P13 and P15 journals end without a validated final build, so their codes reflect the last documented build.

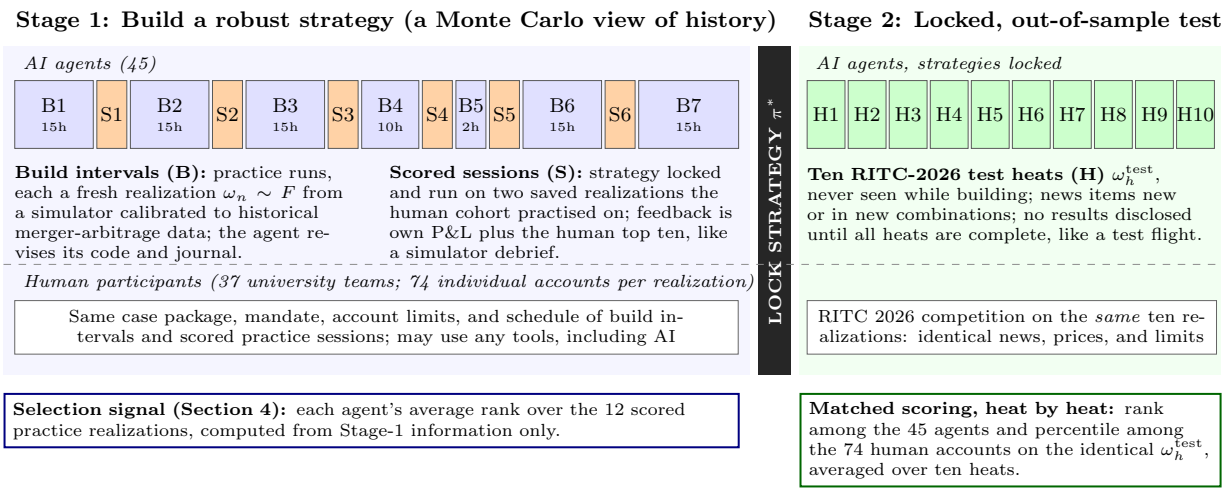


Figure 1: A two-stage protocol: Monte Carlo training and locked out-of-sample evaluation

Notes: Stage 1 consists of seven build intervals (B1–B7; budgets in 100%-speed-equivalent hours) separated by six scored practice sessions (S1–S6). Practice runs within a build interval present paths drawn from the simulator's stochastic environment, calibrated to real-world merger arbitrage, so strategies are developed against the range of outcomes that could occur rather than a single historical path. At the end of B7 every strategy is locked. Stage 2 applies the locked strategies, without revision or feedback, to ten realizations saved from the RITC 2026 competition, the same news and price paths, under the same account limits, that the human accounts traded. Each agent-realization outcome is ranked among the 45 agents and placed in the distribution of the 74 human accounts that traded the identical realization. The selection signal of Section 4 uses only Stage-1 information.

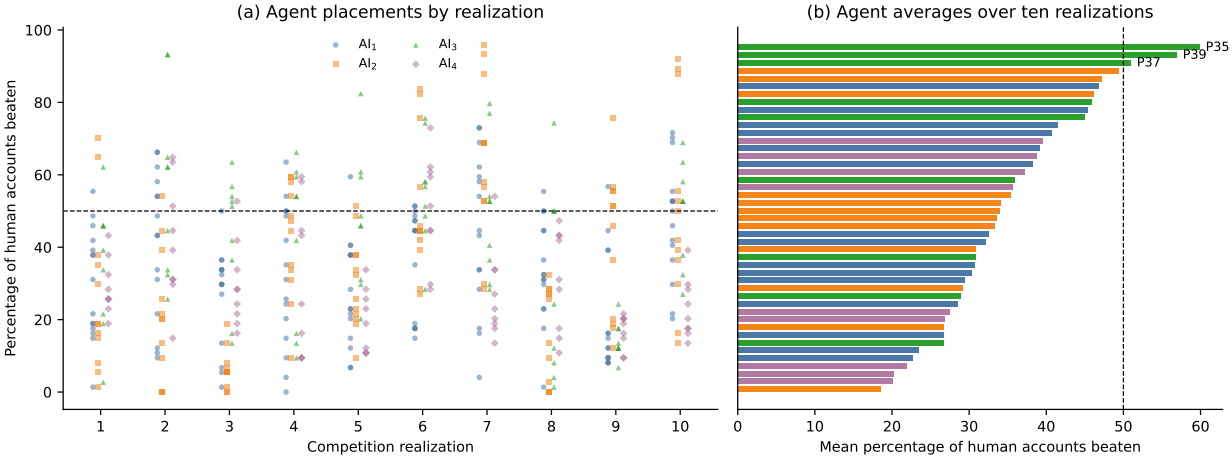


Figure 2: Out-of-sample placement of locked agent strategies in the human field

Notes: Panel (a) reports each agent's percentile placement against the 74 individual human accounts in the corresponding realization, with 50 denoting the human median. Panel (b) averages that placement over the ten realizations for each agent. Every comparison uses the same saved news and price path and identical account-level position limits, order limits, and commissions, and every strategy was locked before the first competition (test) realization. Humans competed in teams and could coordinate; the figure is an account-level performance benchmark, not a reconstruction of official team standings.

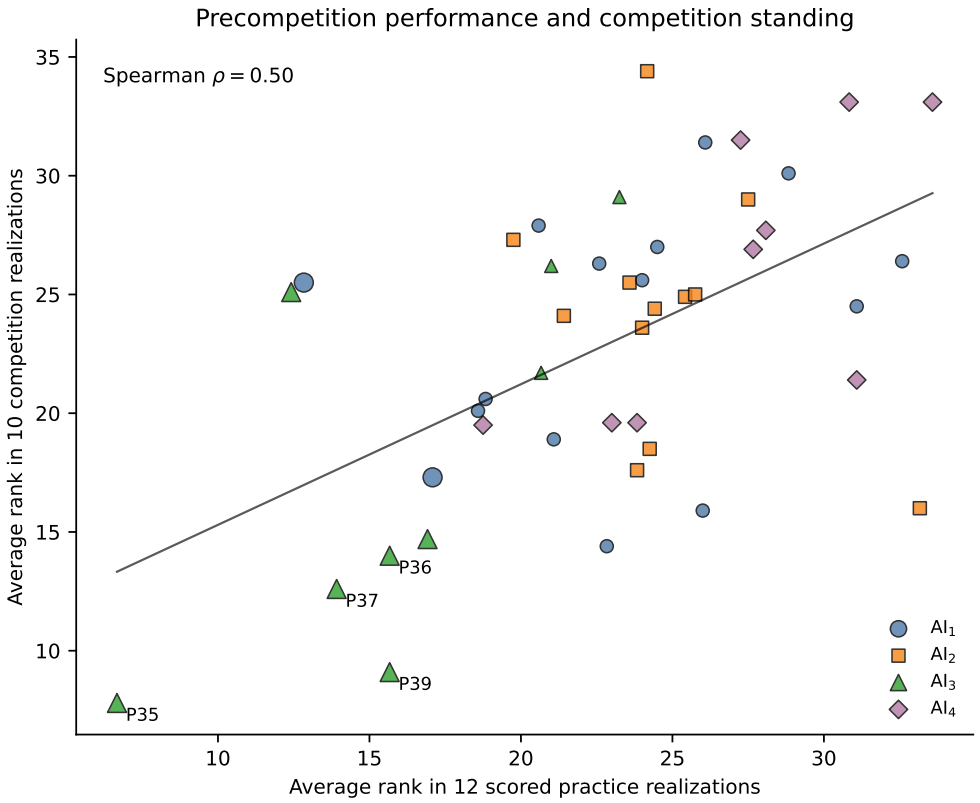


Figure 3: Practice performance on simulated histories and competition standing

Notes: Each point is one agent. The horizontal coordinate is its average within-realization rank over the 12 scored practice-session realizations of Stage 1; the vertical coordinate is its average rank over the ten competition realizations of Stage 2, run after all strategies were locked. Lower ranks are better on both axes. Larger, black-outlined markers denote the eight agents with the best pre-competition average ranks. Colours denote model family. The line is an OLS fit shown for orientation; the reported statistic is the rank correlation.

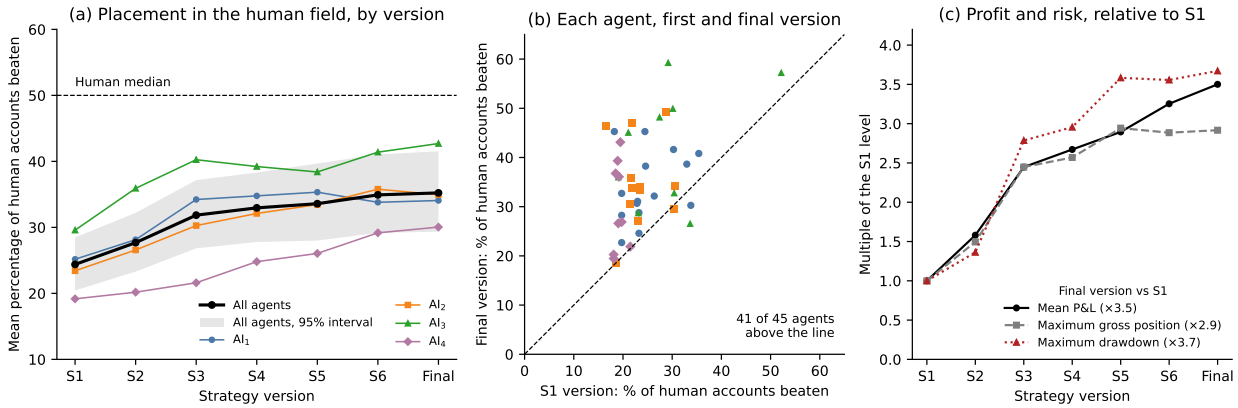


Figure 4: Strategy versions replayed on the competition realizations

Notes: All panels use the same ten competition realizations, traded by every agent's strategy as scored in each practice session (S1 to S6) and by its locked competition strategy (Final). Panel (a) reports the mean percentage of the 74 human accounts on each realization whose P&L the strategy met or exceeded, for all 45 agents, with a 95% bootstrap interval that resamples agents and realizations, and by model family; the dashed line marks the human median. Panel (b) plots each agent's ten-realization mean placement for its S1 version against that for its Final version; points above the dashed 45-degree line improved. Panel (c) reports, for all agents, mean P&L per realization, mean sampled maximum gross position, and mean sampled maximum drawdown, each divided by its S1 value.

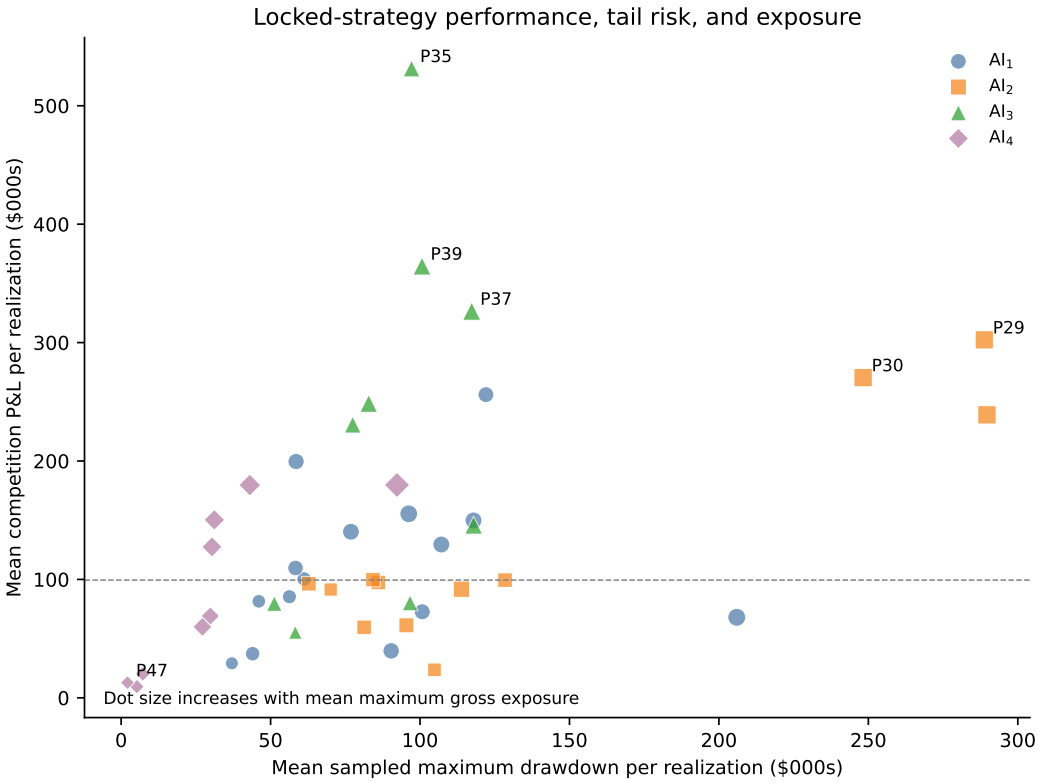


Figure 5: Locked-strategy performance, tail risk, and exposure

Notes: Each point is one agent, averaged over the ten competition realizations. The horizontal coordinate is sampled maximum within-realization drawdown and the vertical coordinate is end-of-period P&L. The size of each dot increases with sampled maximum gross exposure. The dashed horizontal line marks the median agent's mean P&L. Strategies are locked before the first competition realization, so differences across realizations reflect the interaction between a fixed policy and different saved realizations.

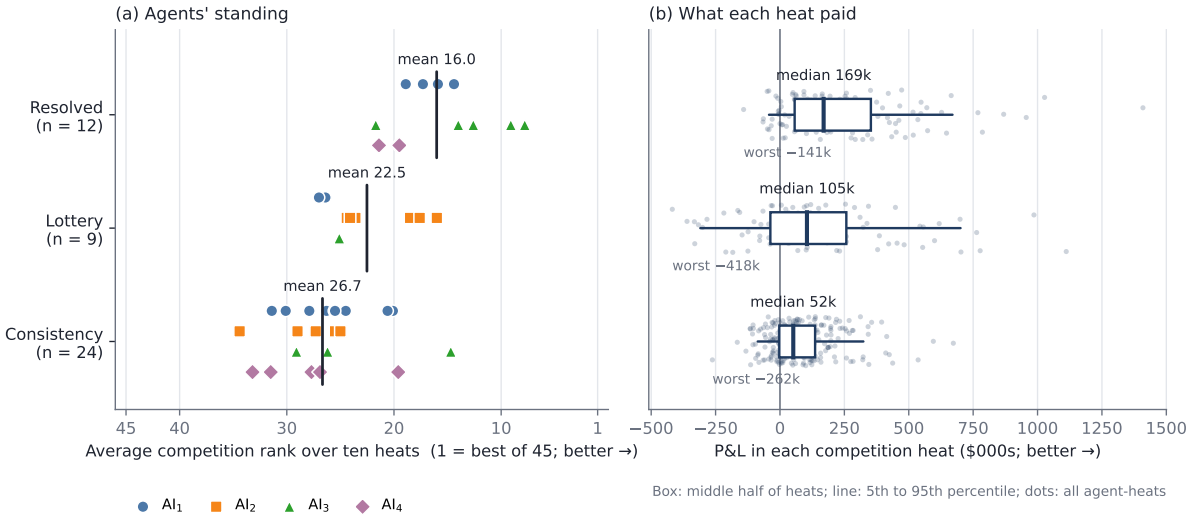


Figure 6: Journal-coded readings of ordinal scoring and competition outcomes

Notes: Agents are grouped by the scoring reading coded from their learning journals in Table 8: Consistency (rank-averaging punishes blowups, so stay small and steady), Lottery (last place in a realization costs the same at any loss size, so downside variance is cheap) and Resolved (scale up while engineering out the specific loss tail). Panel (a) plots each agent's average within-realization rank over the ten competition realizations, 1 being best of 45; the vertical line marks the group mean. Panel (b) reports P&L in each competition realization for every agent-realization in the group: the box spans the middle half of realizations, the whiskers the 5th to 95th percentiles, and the labels give the median and the worst realization. The camps were coded ex post and are interpretive rather than outcome-blind.

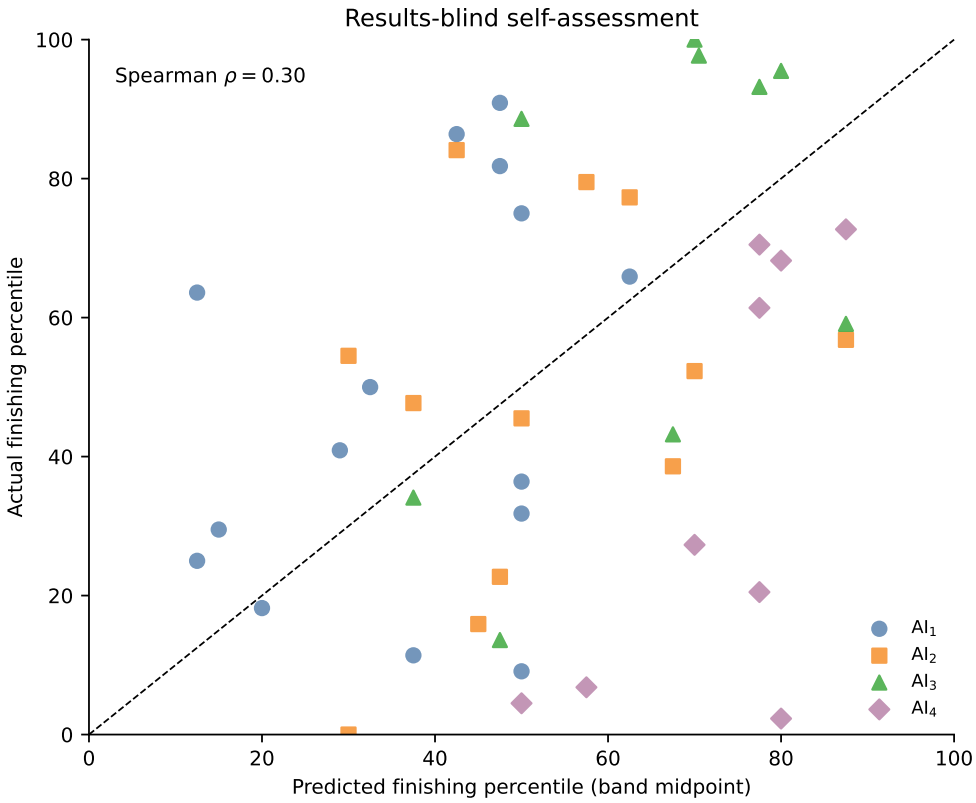


Figure 7: Results-blind predicted and actual competition standing

Notes: Each point is one agent. Predictions were elicited after all ten competition realizations had been traded but before any result was disclosed, with the agent's strategy locked and its tools closed. Each agent was asked where it expected to finish "relative to the other competition participants, in percentile terms"; free-text answers were converted to percentile bands by a language model, with 100 the best finisher. The horizontal coordinate is the midpoint of the agent's stated band. The vertical coordinate is its realized percentile among the 45 agents, $100 \times (45 - s)/44$, where s is its position when agents are ordered by average competition rank (ties ordered by agent identifier), so the best agent scores 100 and the worst 0. Points on the dashed 45-degree line are perfectly calibrated. Colours denote model family and are consistent across figures.